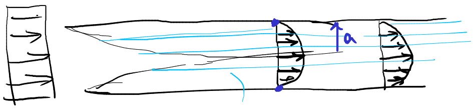

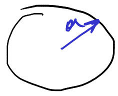

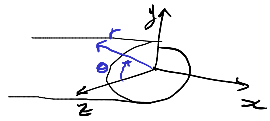

Flujo desarrollado
$$\begin{cases} u = u(y,z): & (\text{no depende } x) \\ \dfrac{Du}{Dt} = 0 & \left(\dfrac{\partial u}{\partial t} = 0 \wedge \dfrac{\partial u}{\partial x} = 0 \right) \end{cases}$$

Como es un problema simétrico también se puede escribir $u(r)$ y v y w son nulos

Pág 232 del libro: $\rho \dfrac{Dv_i}{Dt} = \rho X_i - \dfrac{\partial p}{\partial x_i} + \mu \dfrac{\partial^2 v_i}{\partial x_k \partial x_k}$. (Ec. de Navier-Stokes para fluidos incompresibles)

flujo desarrollado

se desprecia el peso propio

$$\underline{v} = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

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Ec. de N-S en coordenadas cilíndricas

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + \frac{v}{r}\frac{\partial u}{\partial \theta} + w\frac{\partial u}{\partial z} - \frac{v^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r} + \nu\left(\nabla^2 u - \frac{u}{r^2} - \frac{2}{r^2}\frac{\partial v}{\partial \theta}\right) + F_r$$
$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial r} + \frac{v}{r}\frac{\partial v}{\partial \theta} + w\frac{\partial v}{\partial z} + \frac{uv}{r} = -\frac{1}{\rho}\frac{1}{r}\frac{\partial p}{\partial \theta} + \nu\left(\nabla^2 v + \frac{2}{r^2}\frac{\partial u}{\partial \theta} - \frac{v}{r^2}\right) + F_\theta$$
$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial r} + \frac{v}{r}\frac{\partial w}{\partial \theta} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\nabla^2 w + F_z$$

$$\underbrace{\frac{Du}{Dt}}_{0} = \cancel{X} - \frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\nabla^2 u, \Rightarrow \nu \cdot \nabla^2 u = \frac{1}{\rho}\frac{\partial p(x)}{\partial x}$$
$$\cancel{\frac{Dv}{Dt}}^{0} = \cancel{Y} - \frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\nabla^2 \cancel{v}^{0}, \Rightarrow \frac{\partial p}{\partial y} = 0 \Rightarrow p \text{ cte en } y$$
$$\cancel{\frac{Dw}{Dt}}^{0} = \cancel{Z} - \frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\nabla^2 \cancel{w}^{0}, \Rightarrow \frac{\partial p}{\partial z} = 0 \Rightarrow p \text{ cte en } x$$

p solo depende de x

viscosidad cinemática

$$\nu \cdot \nabla^2 u = \frac{1}{\rho} \cdot \frac{\partial p}{\partial x}$$

coeficiente de la viscosidad

$$\frac{\mu}{\cancel{\rho}}\left(\cancel{\frac{\partial^2 u(y,z)}{\partial x^2}}^{0} + \underbrace{\frac{\partial^2 u(y,z)}{\partial y^2} + \frac{\partial^2 u(y,z)}{\partial z^2}}_{f(y,z) = f(r)}\right) = \frac{1}{\cancel{\rho}} \cdot \frac{\partial p(x)}{\partial x}$$

fluido homogéneo

no depende de x

$$\frac{\partial}{\partial x}\left(\mu \cdot f(y,z) = \frac{\partial p(x)}{\partial x}\right)$$

cte de integración

$$\frac{\partial}{\partial x}\left(\frac{\partial p(x)}{\partial x}\right) = 0 \Rightarrow \frac{\partial p(x)}{\partial x} = -\alpha$$

$$\frac{\mu}{\cancel{\rho}}\nabla^2 u(y,z) = \frac{1}{\cancel{\rho}}\overbrace{\frac{\partial p}{\partial x}}^{-\alpha} \Rightarrow \nabla^2 u(y,z) = \frac{-\alpha}{\mu}$$

$$\nabla^2 u(y,z) = \frac{-\alpha}{\mu}$$

Cambiando a coordenadas cilíndricas:

$$\nabla^2 u(r) = \frac{-\alpha}{\mu}$$

Se escribe el Laplaciano en coordenadas polares

Flujo desarrollado — Por la simetría

$$r>0 \qquad \cancel{\frac{\partial^2 u(r)}{\partial x^2}}^{\,0} + \frac{1}{r}\frac{\partial}{\partial r}\left(r.\frac{\partial u(r)}{\partial r}\right) + \frac{1}{r^2}\cancel{\frac{\partial^2 u(r)}{\partial \theta^2}}^{\,0} = -\frac{\alpha}{\mu}$$

$$\int_r \frac{\partial}{\partial r}\left(r.\frac{\partial u(r)}{\partial r}\right) = \int_r -\frac{\alpha r}{\mu} \qquad \text{se integra}$$

$$r.\frac{\partial u(r)}{\partial r} = -\frac{\alpha . r^2}{2\mu} + A \qquad \text{de integración}$$

$$\frac{\partial u(r)}{\partial r} = -\frac{\alpha}{2\mu} r + \frac{A}{r} \qquad \text{Se integra}$$

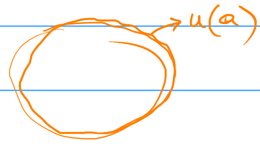
$$u(r) = -\frac{\alpha}{4\mu} r^2 + A . \ln r + B \qquad \text{de integración}$$

Condiciones de borde

- Solución acotada en el dominio (no hay singularidad) por $\ln r \to \infty$ cuando $r \to 0^+$

$\Rightarrow A = 0$

- Condición de no-slip $\quad u(a) = 0 \Rightarrow$

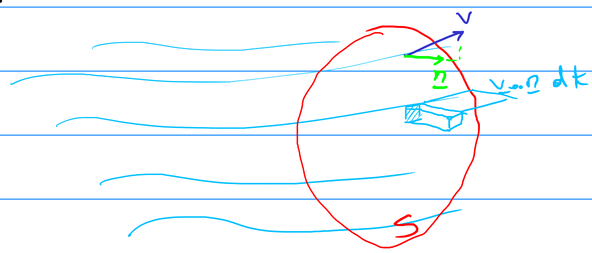


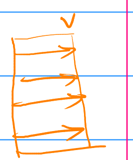
$$u(a) = -\frac{\alpha}{4\mu} a^2 + B = 0 \Rightarrow B = \frac{\alpha a^2}{4\mu}$$

$$\nabla^2 u(r) \to \alpha = -\mu \nabla^2 \mu \to$$
$$\to \frac{\partial P}{\partial x} = -\alpha$$

Reemplazando $\quad u(r) = \frac{\alpha}{4\mu}(a^2 - r^2)$

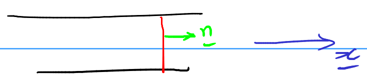
a) Caudal o tasa de flujo de masa:



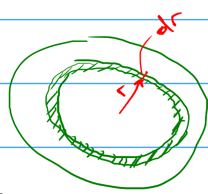


$Q = v\,A\,\rho$

$$Q = \int_S \rho . \underbrace{\underline{v} . \underline{n}}_{h} \; \underbrace{dS}_{base}$$



$$Q = \int_S \rho . \begin{bmatrix} u(r) \\ v \\ w \end{bmatrix} . \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} dS = \int_S \rho . u(r)\, dS \qquad v, w = 0$$



$$Q = \int_{r=0}^{r=a} \rho\, u(r)\, 2\pi r\, dr = 2\pi\rho . \int_0^a \frac{\alpha}{4\mu}(a^2 - r^2)\, r\, dr$$

$$= \frac{\pi}{2}\frac{\rho . \alpha}{\mu}\left(a^2\frac{a^2}{2} - \frac{a^4}{4}\right) = \frac{\pi}{8}\underbrace{\frac{\rho . \alpha}{\mu}}\, . a^4 = -\frac{\pi}{8} a^4 . \rho . \nabla^2 u$$

$$\frac{\alpha}{\mu} = -\nabla^2 u$$